

A clearing fuel market for AeroMAPS

Step 1: formulation, integration, verification and policy results

Branch `feat/fuel-clearing-step1`

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Abstract

AeroMAPS is currently *told* the share of each fuel and computes the resulting cost. This work inverts that: a convex optimisation programme is given a policy — an obligation, a penalty, growth limits, capacities — and returns both the volume of each fuel produced and the price at which it clears. The programme is solved once over all regions, all pathways and all prospective years, so it anticipates: it builds ahead of an obligation it can see coming.

This report states the model in full, describes every parameter, and separates two classes of result that are easy to confuse: those obtained from the optimisation kernel *alone*, at a demand fixed from outside, and those obtained with the kernel *chained into AeroMAPS*, where the price it sets feeds airline costs, airfares and therefore back into demand. It reports what was verified, what was found to be broken along the way, and what the mode cannot yet do.

The headline results: with scarcity switched off the new mode reproduces the existing one to 1.9×10^{-9} across the whole chain; with scarcity on it produces a quantity AeroMAPS has never had — a scarcity rent.

That rent turned out to be exactly what made the coupled fixed point hard to reach, and chasing it down is the other half of this report. The market's price is a Lagrange multiplier; a multiplier is a subgradient of a piecewise-smooth value function; and where that function kinks the price is an *interval* rather than a number, which no fixed-point iteration can be asked to converge on. Giving the energy balance a demand slope — quantity from supply, price from demand, as in any capacity-constrained market — makes the price single-valued again, and is constructed so that it is exactly inert once the loop settles. The coupled model now converges at full marginal pricing, in 60 sweeps and with no acceleration, to a solution of the unmodified programme.

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1 What this is, and what it answers

The existing AeroMAPS energy chain answers: *given these fuel shares, what does the scenario cost?* It cannot answer *given this policy, what fuel gets made?*, because the quantity produced in response to a policy is exactly what a market determines and the current model is handed it as an input.

The fuel market adds a second mode, opt-in and default off. Both modes coexist; they answer different questions. The new one takes an obligation, a penalty for missing it, a production cost and a set of physical limits per pathway, and returns:

- **volumes** — how much of each fuel is produced, in each region, each year;
- **prices** — what the energy is worth at the margin, what the obligation costs to comply with, and what a growth limit is worth relaxing.

The prices are not assumed or calibrated. They are the Lagrange multipliers of the programme's constraints, which is what makes convexity worth paying for: a convex programme has one optimum, and its multipliers are certified prices that do not depend on where the solver started.

2 The model

2.1 The programme

Over regions $r \in \{1..R\}$, pathways $p \in \{1..P\}$ and prospective years $t \in \{1..T\}$, with decision variables $q_{r,p,t} \geq 0$ (energy produced, MJ) and $x_{r,t} \geq 0$ (obligation released by paying the penalty, MJ):

$$\min_{q, x} \sum_t \delta_t \left[\underbrace{\sum_{r,p} \left(c_{r,p,t} q_{r,p,t} + \frac{c_{r,p,t} \gamma_{r,p} K_{r,p,t}}{n+1} \left(\frac{q_{r,p,t}}{K_{r,p,t}} \right)^{n+1} \right)}_{\text{production cost, integral of the marginal cost}} + \sum_r B_{r,t} x_{r,t} \right] \quad (1)$$

subject to, for every region r and year t :

$$\sum_p q_{r,p,t} = D_{r,t} \quad \text{energy balance} \quad \rightarrow \lambda_{r,t}^E \quad (2)$$

$$\sum_{p \in \mathcal{S}_r} q_{r,p,t} + x_{r,t} \geq m_{r,t} D_{r,t} \quad \text{obligation} \quad \rightarrow \lambda_{r,t}^M \quad (3)$$

$$\sum_{p \in \mathcal{N}_r} q_{r,p,t} + x_{r,t}^{\text{sub}} \geq \mu_{r,t} D_{r,t} \quad \text{sub-obligation} \quad \rightarrow \lambda_{r,t}^S \quad (4)$$

$$q_{r,p,t} \leq s_{r,p} + (1 + g_{r,p}) q_{r,p,t-1} \quad \text{growth limit} \quad \rightarrow \lambda_{r,p,t}^R \quad (5)$$

with $q_{r,p,0} \equiv q_{r,p}^{\text{init}}$ given. $\mathcal{S}_r$ is the set of pathways *eligible* for region r 's obligation and $\mathcal{N}_r \subseteq \mathcal{S}_r$ the narrower set for its sub-obligation.

Three properties of this programme carry the design.

One solve, not one per year. Constraint (5) ties t to $t-1$, so solving year by year would discard anticipation. Sustainable output above the obligation in early years is a *result*, not an artefact: with perfect foresight the cheapest way to meet a step is to be ready before it arrives.

Convex, so the multipliers are prices. λ^E is the marginal cost of energy; λ^M is the compliance price. A fixed point found by iteration would not come with that guarantee.

The growth limit is a sum, not a maximum (§2.6).

2.2 Lagrange multipliers, and the exact sense in which they are prices

The claim in the box above — “ λ^E is the marginal cost of energy” — is the load-bearing one in this whole design, and it is worth deriving rather than asserting, because the same derivation shows precisely where it stops holding.

The Lagrangian. Write the objective (1) as $C(q, x)$ and attach a multiplier to each constraint, signed so that every multiplier is non-negative on an inequality:

$$\begin{aligned} \mathcal{L}(q, x; \lambda) = & C(q, x) + \sum_{r,t} \lambda_{r,t}^E (D_{r,t} - \sum_p q_{r,p,t}) \\ & + \sum_{r,t} \lambda_{r,t}^M (m_{r,t} D_{r,t} - \sum_{p \in \mathcal{S}_r} q_{r,p,t} - x_{r,t}) \\ & + \sum_{r,p,t} \lambda_{r,p,t}^R (q_{r,p,t} - s_{r,p} - (1 + g_{r,p}) q_{r,p,t-1}). \end{aligned} \quad (6)$$

Stationarity is a statement about margins. Setting $\partial \mathcal{L} / \partial q_{r,p,t} = 0$ for a pathway produced in strictly positive quantity gives, after dividing through by the discount factor δ_t :

$$\underbrace{c_{r,p,t} \left(1 + \gamma_{r,p} \left(\frac{q_{r,p,t}}{K_{r,p,t}} \right)^n \right)}_{\text{marginal production cost}} + \underbrace{\lambda_{r,p,t}^R - (1 + g_{r,p}) \frac{\delta_{t+1}}{\delta_t} \lambda_{r,p,t+1}^R}_{\text{scarcity of the right to grow}} = \lambda_{r,t}^E + \lambda_{r,t}^M \mathbf{1}[p \in \mathcal{S}_r]. \quad (7)$$

Read from the right: what a unit of this fuel is *worth* to the system — the energy it supplies, plus the obligation it discharges if it is eligible — equals what it costs at the margin, plus the shadow value of the growth room it consumes. That is an identity, not a modelling assumption. Three things follow from it directly.

- **Equation (7) is why the ramp-up multiplier never appears in any published price.** It is already inside λ^E and λ^M . A model that added λ^R to the price the airline pays would count it twice.
- **A pathway not produced** has $\partial\mathcal{L}/\partial q \geq 0$ instead: it is worth less than it costs. The programme's choice of which fuels to build is exactly this comparison.
- **The term $-(1+g)\lambda_{t+1}^R$ is anticipation, written down.** Growth room used this year buys growth room next year. Its sign is negative: building now is *worth* something later. This single term is why the solve builds ahead of an obligation (§5), and it exists only because all years are solved at once.

Complementary slackness. For every inequality, $\lambda \geq 0$, the slack is ≥ 0 , and their product is zero:

$$\lambda_{r,t}^M \cdot \left(\sum_{p \in \mathcal{S}_r} q_{r,p,t} + x_{r,t} - m_{r,t} D_{r,t} \right) = 0. \quad (8)$$

An obligation that is over-met has *no* compliance price; one that is exactly met has one. This is a testable property rather than an interpretation, and it is what the parametrised test over seven regimes checks (§4). Every degeneracy found in this work was found by it failing.

The envelope theorem is the actual claim. Let $V(D)$ be the optimal value of the programme as a function of the demand vector. Then at an optimum

$$\lambda_{r,t}^E = \frac{\partial V}{\partial D_{r,t}}. \quad (9)$$

The multiplier is the derivative of the cost of the whole system with respect to asking it for one more MJ — every knock-on effect included, across pathways, across years, through the ramp-up chain. *That* is what makes it a price and not a bookkeeping quantity, and it is unavailable from any allocation rule: a rule can report what each fuel cost, but only an optimum can report what the next unit would cost.

Why convexity is not a technicality. Equation (9) needs strong duality, which for this programme follows from Slater's condition (the feasible set has a relative interior whenever the obligation is reachable, and the buy-out x guarantees that it is). Convexity also delivers what no iterative scheme can: the optimum is global and the duals are certified, so the answer does not depend on where the solver started. This is the entire reason the market is a dedicated convex solve rather than another block inside the MDA (§3.3).

The caveat that the rest of this report is about. Equation (9) is an equality only where V is differentiable. In general the correct statement is

$$\lambda \in \partial V(D) \quad (\text{the subdifferential}),$$

and V is piecewise smooth, not smooth: it has a kink wherever the set of tight constraints changes. At such a point ∂V is an *interval*, every member of it satisfies the KKT conditions, and the solver returns one of them with no way to prefer another. The price is genuinely not determined by the cost side there.

Nothing is wrong with the optimisation. But a quantity that is set-valued cannot be the input to a fixed-point iteration, and that — not stiffness, not scaling, not the solver — is

why the coupled loop failed at $w = 1$ (§6.4), and what §6.5 does about it.

2.3 Every parameter

Symbol	Shape, unit	Meaning and provenance
$D_{r,t}$	(R, T) , MJ	Drop-in energy demand. In coupled mode this is <code>energy_consumption_dropin_fuel</code> , an output of the fleet chain — which is what makes the market part of a loop.
$c_{r,p,t}$	(R, P, T) , EUR/MJ	Production cost per unit energy at zero utilisation. Taken as <code>{p}_net_mfsp</code> : the cost the buyer actually faces, carbon tax and subsidies included (§3.4). Flat in time on the bench — AeroMAPS has no learning curve, so all price dynamics come from the model, not the supply curve.
$m_{r,t}$	(R, T) , fraction	The obligation: minimum eligible share of demand. A <i>fraction</i> , not a percentage — validated, because a silent factor of 100 here would be invisible downstream.
$\mathcal{S}_r$	(R, P) , boolean	Eligibility , not chemistry: which pathways count towards region r 's obligation. The same fuel may count in one jurisdiction and not another.
$\mu_{r,t}, \mathcal{N}_r$	$(R, T), (R, P)$	Sub-obligation and its narrower eligible set — ReFuelEU's synthetic-fuel target. Optional; absent by default.
$B_{r,t}$	(R, T) , EUR/MJ	Penalty (“buy-out”) per unit of unmet obligation. Caps λ^M : nobody pays more to comply than to not comply. Without it an unreachable obligation makes the programme <i>infeasible</i> rather than expensive.
$K_{r,p,t}$	(R, P, T) , MJ	Capacity <i>scale</i> , not a cap: the output at which the pathway starts to strain. ∞ disables saturation. Exogenous at step 1.
$\gamma_{r,p}$	(R, P)	Saturation intensity — how much dearer the next unit is at full utilisation. 0 switches the term off entirely.
n	scalar ≥ 0	Saturation stiffness — how <i>abruptly</i> the extra cost arrives. $n \rightarrow \infty$ recovers a hard cap, which is what the soft form exists to avoid.
$g_{r,p}$	(R, P) , /year	Growth limit: relative expansion allowed year on year.
$s_{r,p}$	(R, P) , MJ	Growth <i>seed</i> . Not optional in practice: with $q^{\text{init}} = 0$ and $s = 0$, (5) pins the pathway at zero forever. Not a free dial either — it is Eq. (12)'s volume branch (§2.6).
$q_{r,p}^{\text{init}}$	(R, P) , MJ	Production in the last historical year; the growth limit's initial condition.
δ_t	scalar	Discount factor $(1 + \rho)^{-t}$. Decides how far ahead of a step it is worth building.
w	scalar $\in [0, 1]$	Pricing weight . Decides who keeps the scarcity rent (§2.5). The single most consequential parameter.

Table 1: Inputs. R regions, P pathways, T prospective years.

2.4 The prices the model produces

The *marginal price* of a pathway aggregates the constraints one unit of it relaxes:

$$\text{MC}_{r,p,t} = \lambda_{r,t}^E + \lambda_{r,t}^M \mathbf{1}[p \in \mathcal{S}_r] + \lambda_{r,t}^S \mathbf{1}[p \in \mathcal{N}_r]. \quad (10)$$

Symbol	From	Reads as
$\lambda_{r,t}^E$	(2)	What the next unit of energy costs. Set by whichever pathway is on the margin.
$\lambda_{r,t}^M$	(3)	What one more unit of obligation costs. Bounded above by B . Zero in any year where the market is already ahead of the obligation — a slack constraint costs nothing.
$\lambda_{r,t}^S$	(4)	The same, for the narrower target.
$\lambda_{r,p,t}^R$	(5)	What it would be worth to be allowed to grow faster. Zero where the limit is not in the way.

A sub-mandated unit carries both obligation multipliers. That is not double counting: it genuinely relaxes two distinct constraints.

2.5 What the airline actually pays

This section answers a question the earlier write-ups left implicit.

Two prices exist for each pathway and they are not the same. The **average cost** is what the fuel produced actually cost, per unit:

$$AC_{r,p,t} = c_{r,p,t} \left(1 + \frac{\gamma_{r,p}}{n+1} \left(\frac{q_{r,p,t}}{K_{r,p,t}} \right)^n \right), \quad (11)$$

against the marginal price (10). The gap between them is **inframarginal rent** — profit earned on every unit produced more cheaply than the last one. The pricing weight interpolates:

$$\text{mfsp}_{r,p,t}^{\text{mkt}} = (1-w) AC_{r,p,t} + w MC_{r,p,t} - \underbrace{\left(\text{net}_{r,p,t} - \text{gross}_{r,p,t} \right)}_{\text{tax wedge, §3.4}} \quad (12)$$

and AeroMAPS’s existing machinery then weights that across pathways into the one number the airline sees, $\{\text{type}\}_{\text{mean_mfsp}} = \sum_p \text{share}_p \cdot \text{mfsp}_p^{\text{mkt}}$, which drives operating cost, airfare and demand.

So: fuel cost, or compliance, or both? The honest answer is *both, always* — but not by addition, and this is the easiest thing in the model to get wrong.

Region A, 2050	Airline pays	λ^M	SAF share
No obligation at all	0.01200	0	0 %
70 % obligation, $w = 0$	0.01982	0.01117	70 %
70 % obligation, $w = 1$	0.01982	0.01117	70 %
+ scarcity, $w = 0$	0.02115	0.07917	70 %
+ scarcity, $w = 1$	0.06742	0.07917	70 %

The obligation is paid for at $w = 0$: the bill goes from 0.01200 to 0.01982, a **65 % increase**. It is paid through the *mix* — the obligation forces 70 % of the fuel to be the expensive kind — not through a separate charge. The cost of compliance is *embedded*, not itemised.

λ^M is not added on top of that, and must not be. It is the *marginal* cost of the obligation — what one more unit of it would cost. The average cost increment is $0.01982 - 0.01200 = 0.00782$, which is exactly $0.70 \times \lambda^M$: share times marginal cost. Adding λ^M to the delivered price would give 0.03099, a 158 % increase, **charging the obligation twice**.

What w changes is average versus marginal, not paid versus unpaid. At $w = 1$ every unit is priced at the *last* unit's cost. With nothing scarce that is the same number (rows 2 and 3 are identical — marginal equals average when the supply curve is flat). With scarcity it is not: 0.06742 against 0.02115, and the difference is inframarginal rent.

It is never “compliance only”. λ^M is a premium on top of the energy price, never a substitute for it.

The carbon tax arrives separately, as its own `DirectOperatingCosts` line — which is exactly why (12) subtracts the tax wedge back out (§3.4). Same double-counting logic, different instrument.

One thing the airline does *not* pay, and arguably should. The buy-out payment Bx appears in objective (1) but in no price output. If the market releases part of the obligation, that penalty is borne by nobody in the cost chain. At $w = 1$ the *rate* is signalled correctly (λ^M is capped at B), but the *payment* is not accounted anywhere. This is a known gap, not a modelling choice.

A useful consistency check falls out: with no scarcity at all, $\lambda^E = c_k$ (kerosene) and $\lambda^M = c_s - c_k$, so a sustainable pathway's marginal price is exactly c_s — which is also its average cost. The two endpoints of (12) coincide and w is inert. w **only matters where something is scarce**, from *either* the capacity term or a binding growth limit.

2.6 Two choices that keep the problem convex

Capacity squeezes, it does not wall. Production is not capped. Instead the marginal cost rises with utilisation, $c(1 + \gamma(q/K)^n)$, whose integral appears in (1). A hard cap would make the price jump discontinuously at $q = K$; since that price feeds airfares, which feed demand, which feeds fuel demand, a jumping price makes the coupled loop oscillate instead of settle. Figure 1 shows the shape and the rent.

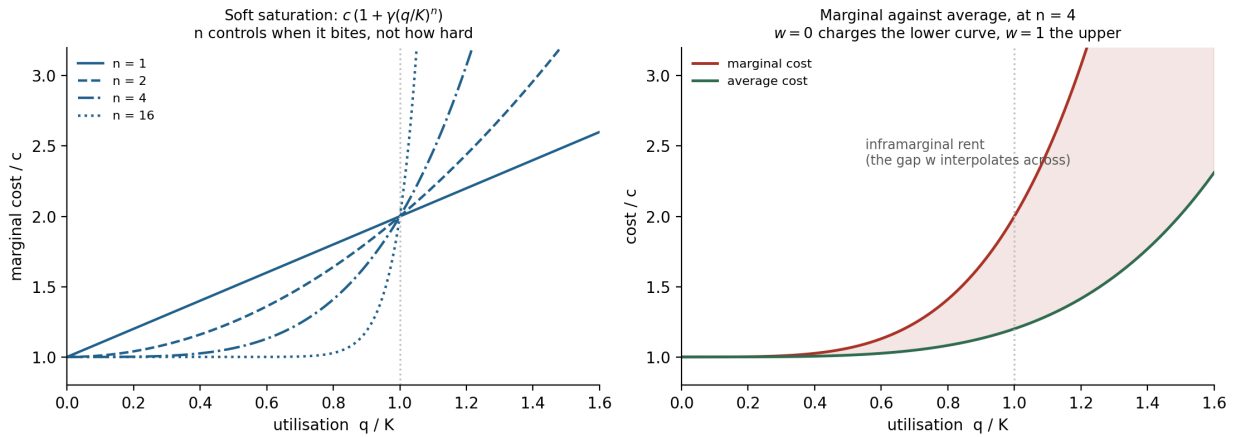


Figure 1: Left: n decides *when* the squeeze bites, not how hard. Note the consequence — below $q = K$ a *larger* n gives a *smaller* markup, which is why stiffness reduces the price on a bench running under capacity. Right: the gap between marginal and average cost is the rent that w interpolates across.

The growth limit has to be a sum. AeroMAPS's optimisation mode limits growth by Eq. (12) of the published work — the *more permissive* of a relative rate and a fixed volume addition:

$$E_t \leq \max\{(1 + \tau) E_{t-1}, E_{t-1} + \Delta E\}. \quad (13)$$

The region under a maximum of two affine functions is not a convex set (Figure 2, right), so (13) cannot be handed to a convex solver. This is not a tolerance problem; the feasible set is the wrong shape. Constraint (5) replaces the maximum with a *sum*, which is its convex outer relaxation: it admits everything (13) admits, plus a margin **measured at 0.73 %, independent of g** .

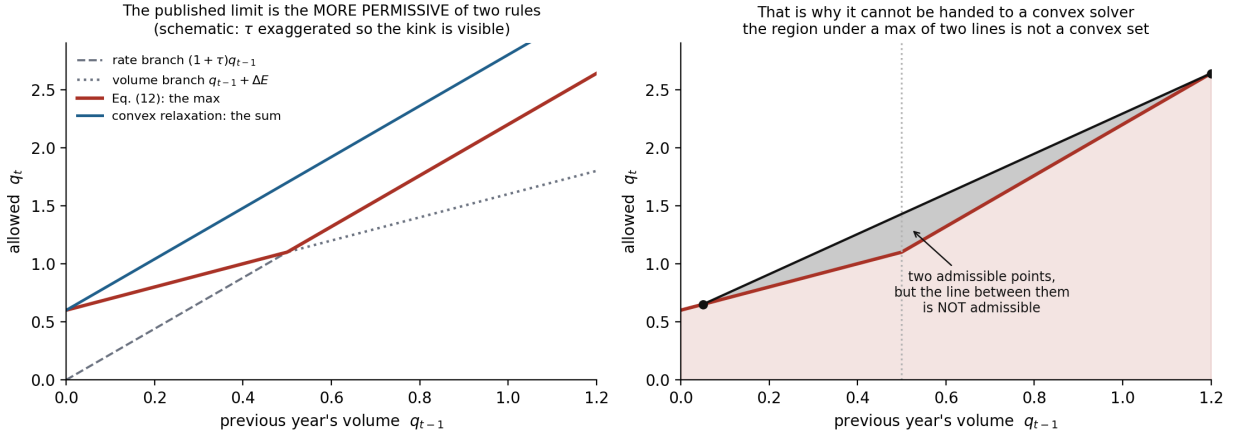


Figure 2: Why the published growth limit cannot go into the programme, and what replaces it. Right: two admissible points whose connecting line is inadmissible — the definition of a non-convex set.

The seed s in (5) is Eq. (13)’s volume branch ΔE , so it is *not* a free parameter: the published calibration (0.2 EJ/yr at an ASK share of 0.1549) puts it at about **1.5 % of annual demand**. It matters — on the bench the seed moves the compliance price by a factor 4.3 across a plausible range.

3 How it plugs into AeroMAPS

3.1 Two levels of result, and why the distinction matters

This section answers the second question the earlier write-ups left implicit.

Results in this report come from one of two set-ups, and conflating them would be a serious misreading.

	Kernel only	Chained into AeroMAPS
What runs	The convex programme alone. NumPy arrays in, arrays out. No AeroMAPS import.	The programme as a GEMSEO discipline inside the full multi-regional MDA.
Demand $D_{r,t}$	Fixed from outside — taken from a stored reference run.	An output of the model , responding to the price the market sets.
Loop closed?	No. Price has no effect on demand.	Yes: price $\rightarrow$ operating cost $\rightarrow$ airfare $\rightarrow$ elasticity $\rightarrow$ demand $\rightarrow$ price.
Cost per case	~ 0.03 s	minutes
Used for	Formulation, verification, parameter sweeps, all policy cases .	Reproduction end-to-end, convergence, the coupled grid.

Table 2: The two levels. Everything in §5 and §7 is kernel-only; everything in §6 is coupled.

Why the policy cases are kernel-only, deliberately. Chaining scales every price effect by roughly a tenth (§6.3) and changes no *ordering* at $w = 0$, while costing minutes per case. (At $w > 0$ it also used to introduce a convergence problem, §6.4; that is now resolved, §6.5, but the cost per case remains.) The kernel-level results are therefore the *policy* answer; the coupled results are what tells you how much of it survives the demand response.

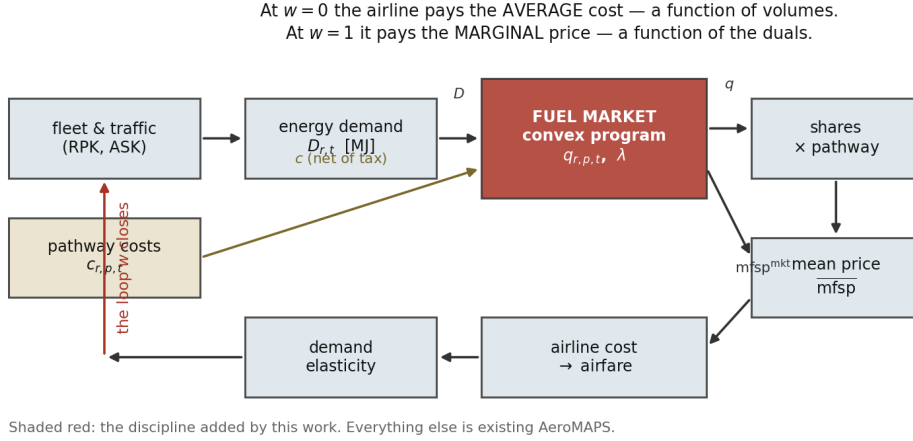


Figure 3: Where the market sits. Shaded: the discipline added by this work. The loop on the left is closed only in coupled mode, and only carries a discontinuous quantity when $w > 0$.

3.2 What the discipline replaces

In market mode the existing allocator `EnergyUseChoice` is **not instantiated**: the market decides the volumes rather than being handed shares. It emits the same nine output families under the same names, computed by a *shared* function extracted from the allocator rather than a second copy — two copies would drift, and the failure is silent, since a share family disagreeing with its volumes rescales CO_2 with no error raised.

The mode is one configuration key, default off, propagated to each region as a process-level keyword. It is refused outside the unified-MDA execution mode, and three guards raise at set-up rather than producing plausible-looking numbers: drop-in fuels only, top-down cost model only, exactly one residual pathway.

3.3 The MDA, and why it is an awkward instrument for this particular coupling

AeroMAPS closes its loops with a Multi-Disciplinary Analysis: GEMSEO orders the disciplines, identifies the strongly coupled block (here 63 variables across two regions), and sweeps it with Gauss–Seidel until a residual norm falls below tolerance. For the couplings AeroMAPS was built around — fleet renewal, energy intensity, traffic — this is exactly right. The market is a worse fit, and it is worth being explicit about why, because the limitation is structural rather than a matter of settings.

An MDA is a fixed-point iteration, and it asks for more than it looks like. Gauss–Seidel converges when the composite map from coupling variables to coupling variables is a contraction near the solution. In this loop that map is

$$p \mapsto D(p) \mapsto \Lambda(D(p)) = p',$$

where $D(\cdot)$ is the whole fleet–cost–airfare–elasticity chain and Λ is the market’s price. Convergence therefore requires Λ to be not merely bounded but *Lipschitz* in demand, with a small enough constant. At $w = 0$ it is: the price is the average cost, an explicit smooth function of the volumes. At $w = 1$ it is a multiplier, and §2.2 established that a multiplier is a subgradient of a piecewise-smooth value function. Its Lipschitz constant is infinite at every kink. No choice of relaxation factor or acceleration method repairs that, because the failure is in the map, not in the scheme applied to it — a point this work initially got wrong and had to retract (§8).

The disciplines exchange point values, and a set-valued quantity has none. The MDA’s contract is that each discipline is a function: same inputs, same outputs. At a kink the market is a *correspondence* — the KKT system admits an interval of prices, and which one comes back is a property of the solver’s trajectory. It is consistent with the maths and inconsistent with the contract. Nothing in GEMSEO can detect this; the discipline looks like it is returning a number, and it is.

The convergence test measures the wrong thing. The residual is a norm over all 63 coupling variables, in their own units. A market that has settled to eleven digits can sit inside a residual dominated by `region_B:ask` at 10^{12} , and a market that is flipping between two active sets can hide inside a residual that is falling for unrelated reasons. Diagnosing §6.4 required instrumenting the discipline itself — and four successive instrumentation attempts were wrong before one was right (§8), each producing a confident, false reading.

The market is solved many times to answer one question. The convex programme is solved *in full* — all regions, pathways and years, with perfect foresight — at every sweep. A converged coupled case costs 23 solves at $w = 0$ and 60 at $w = 1$; a failing one costs 202. Each is a globally optimal answer to a question that is about to be superseded.

What the alternative would be. All of the above points the same way: the demand response belongs *inside* the programme rather than outside it. A single optimisation maximising surplus net of cost, with demand as a decision variable, has no outer loop, no set-valued exchange and no residual to interpret — its optimality conditions simply include the demand relation. That is a large change to how AeroMAPS is organised and it is not proposed here.

What *is* done here (§6.5) is the smallest useful step in that direction: a local linearisation of the demand curve is passed into the programme, enough to make the price single-valued, constructed so that it is exactly inert once the loop settles. The MDA stays; it is simply no longer asked to iterate on a quantity that has no value to iterate on.

3.4 Decide on net, report on gross

The market chooses using `net_mfsp` — cost *including* carbon tax and subsidies — because that is what determines which pathway is marginal. This is not cosmetic: on the bench a carbon tax of only 5 EUR/t already narrows the sustainable premium by 3 %, and extrapolating the emission factors the premium reaches zero near **164 EUR/t**, where the ordering reverses and the market buys sustainable fuel unprompted. A market clearing on gross cost could never produce that.

But `DirectOperatingCosts` adds the carbon tax as its own line. Publishing a net-basis price would count the tax twice — hence the wedge subtracted in (12). **Decide on net, report on gross, and the tax is counted once.**

A *passenger*-level tax is deliberately excluded from the objective: it is uniform across fuels, so it cannot change which pathway is marginal. It still reaches the market through demand, which is the correct channel.

4 Verification

An analytic case pins the conventions. One region, one year, saturation known in closed form, so both prices are known exactly. This caught a sign error that documentation could not have: `cvxpy` reports the equality’s multiplier negated, so the energy price came out as -0.012 against an expected $+0.012$ — a plausible number with the wrong sign.

38 kernel tests , covering the balance, the obligation, the growth limit, scale invariance, continuity and input hygiene.

Complementary slackness, across seven regimes. A multiplier may be non-zero only where its constraint is tight. This single property is what all three real defects (§8) violated, and nothing tested it. It is now parametrised over the *edges* — zero obligation, full obligation, a constraint that cannot bind — because a mid-range case sees none of them. Worst residual: 1.4×10^{-8} , solver noise; the phantom compliance price scored 0.46 on the same measure.

5 Results without the chain (kernel only)

5.1 It reproduces the current allocator

Given the reference run’s shares, with saturation off and the growth limit loose, the market lands on the reference allocation with a residual below 10^{-8} of demand — an order of magnitude under solver noise.

5.2 Perfect foresight builds ahead of the obligation

At $g = 15\%/yr$ against the ReFuelEU steps, region A holds **more** sustainable fuel than it owes in every year between steps: 14.1 % in 2034 against a 6 % obligation, 28.9 % in 2039 against 20 %. The growth limit cannot take it from 6 % to 20 % in the year the step lands, so it starts early. Sustainable fuel is the expensive option and discounting penalises early spend, so it overshoots by the *minimum* that clears the next step — which is what a firm reading a published mandate would do.

The penalty is needed exactly once, at the 2035 step and only at the tightest growth limit. The compliance price is **zero in every year where the market is ahead**.

5.3 Which parameters move the answer

Parameter	Swept	Spread in delivered price
Pricing weight w	$0 \rightarrow 1$	37.3 %
Stiffness n	$1 \rightarrow 16$	16.3 %
Penalty B	$0.02 \rightarrow 0.30$	8.3 %
Growth limit g	$0.10 \rightarrow 1.20 /yr$	3.8 %
Intensity γ	$0 \rightarrow 4$	1.7 %
Discount rate ρ	$0 \rightarrow 0.08$	0.0 %

Two readings that a single-parameter view gets wrong. γ ’s **1.7 % is misleading**: it was measured at one stiffness, and at $n = 1$ the same range moves the uplift $4\% \rightarrow 16\%$. The two saturation parameters are not separable. And γ ’s real effect at this baseline is on the *compliance* price, which it moves $+57\%$ — scarcity raises the cost of the obligation well before it raises the average fuel bill.

The penalty is inert until it is cheap enough to be worth paying. The obligation’s own marginal cost here is 0.0546 EUR/MJ. Set B above that and it never binds; set it below and

the market stops building and pays instead, and the *delivered* price falls 8.3 %. A penalty priced under the cost of compliance is a subsidy to non-compliance.

5.4 Capacity sized on the obligation is undersized

With a binding growth limit, anticipation pushes utilisation to $q/K = 2.01$ between steps: the market builds twice the plant the obligation asks for. Saturation disciplines it — at $\gamma = 1$ the peak falls to 1.30.

6 Results with the chain (coupled)

6.1 End-to-end reproduction

With nothing scarce, the coupled mode must reproduce the existing model. It does, and not only in the volumes:

2050, both regions	reference	market	worst rel., all years
hefa_fog_energy_consumption	9.84171e12	9.84171e12	1.1e-09
hefa_fog_share_dropin_fuel	70	70	1.9e-09
dropin_fuel_mean_mfsp	0.019819	0.019819	1.8e-10
dropin_fuel_mean_co2_emission_factor	41.1	41.1	1.0e-09
co2_emissions_passenger	493.04	493.04	6.3e-10
airfare_per_rpk	0.0867286	0.0867286	5.1e-11
rpk	1.98233e13	1.98233e13	1.6e-11

Table 3: Worst relative difference across every variable and year: 1.9×10^{-9} .

An independent confirmation fell out: the compliance price came back at 0.01083, which is the *net*-basis premium (0.010830) and not the gross one (0.011170). The decide-on-net rule is doing what it claims.

6.2 Scarcity reaches the traffic loop

	no scarcity	scarce, $w = 0$	scarce, $w = 1$
Delivered price 2050	0.019819	0.021839 (+10.2 %)	0.04020 (+103 %)
Max compliance price	0.01083	0.047886	—
RPK 2050	1.98233e13	−1.2 %	−10.8 %
CO ₂ 2050	493.04	−1.2 %	—

Expensive fuel makes tickets dearer and some journeys do not happen. AeroMAPS could not previously produce that chain of consequences *from a fuel policy*, because nothing in it priced scarcity.

6.3 The coupled saturation grid

Swept through the full MDA, so volumes respond. Delivered price % / RPK % against the no-scarcity run:

The anchored run agrees with the rigid one wherever the rigid one worked. This is the system-level check that §6.5’s inertness argument survives contact with the full chain, and it is worth more than the kernel test because it exercises every discipline between the market and the demand:

	$n = 2$	$n = 4$	$n = 8$	$n = 16$
$w = 0, \gamma = 0.5$	+11.2/ - 1.3	+5.2/ - 0.6	+1.8/ - 0.2	+0.4/ - 0.0
$w = 0, \gamma = 1$	+21.9/ - 2.5	+10.2/ - 1.2	+3.6/ - 0.4	+0.8/ - 0.1
$w = 0, \gamma = 2$	+42.1/ - 4.7	+19.6/ - 2.3	+7.1/ - 0.8	+1.6/ - 0.2
$w = 1, \gamma = 0.5$	+99.6/ - 10.5	+84.2/ - 9.0	+71.8/ - 7.8	<i>n.c.</i>
$w = 1, \gamma = 1$	+131.9/ - 13.5	+102.9/ - 10.8	+80.4/ - 8.7	<i>n.c.</i>
$w = 1, \gamma = 2$	+186.2/ - 18.0	+133.2/ - 13.6	+94.7/ - 10.0	<i>n.c.</i>

Table 4: 21 of 24 cells converge with the demand-anchored balance and *no* acceleration — all 12 at $w = 0$ and 9 of 12 at $w = 1$, against 3 of 12 before (§6.5). The three remaining failures are the whole $n = 16$ column, the stiffest saturation on the grid; they are a separate and still-open problem.

- all 13 $w = 0$ cells: agree to $\sim 10^{-10}$ relative — unchanged, as required;
 - the 3 $w = 1$ cells that converged before: agree to 3×10^{-8} ;
 - 6 $w = 1$ cells that previously failed: now converge.
- The demand slope changed nothing that already worked, and fixed what did not.

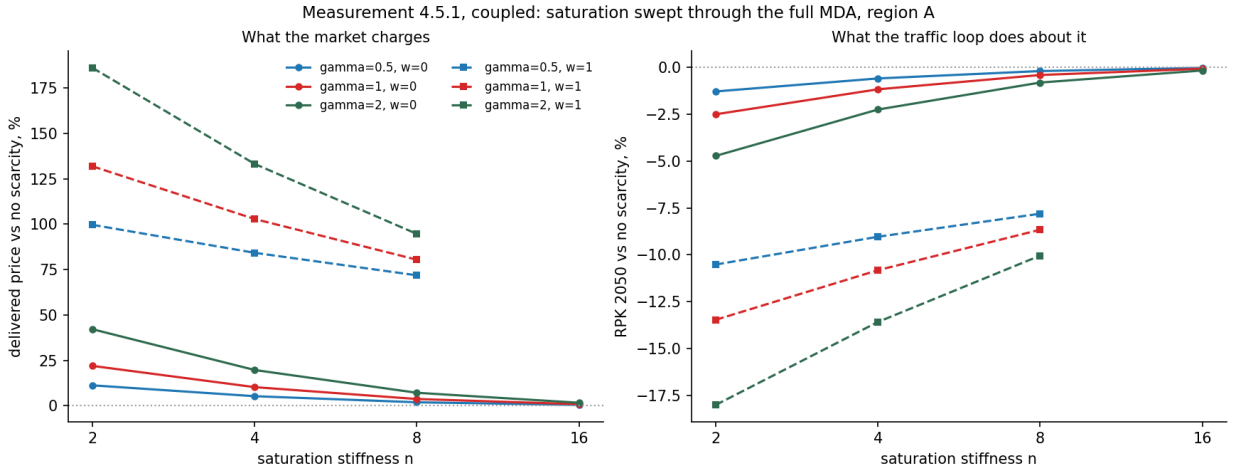


Figure 4: The same grid drawn. Gaps are cells that did not converge; they are not interpolated over.

Two results survive the feedback intact. The kernel-level *ordering* holds — uplift still falls monotonically in n , at $w = 1$ as well now that the row is legible. And **the traffic response is consistently about a tenth of the price response**: +42.1 % on price gives -4.7 % on RPK. That ratio is the product of two independently measured transmissions (fuel is 13.1 % of the airfare; demand elasticity 0.888). Rule of thumb for this bench: **a 10 % fuel-price rise costs about 1.2 % of traffic**.

6.4 The coupled fixed point, and why it failed at $w = 1$

At $w = 1$ the loop does not converge under AeroMAPS’s default settings: the residual stalls at 8.6×10^{-2} after 200 sweeps. Damping helps but does not close it, and acceleration — which an earlier version of this report credited with fixing it — does not (§8).

The mechanism was first *inferred* from the shape of the residual. It has since been *measured*. The kernel now reports an `active_signature`: a digest of which constraints are tight, taken from the primal slacks and never from the multipliers, since whether a multiplier is non-zero is precisely the question that degenerates. Two sweeps with the same signature solved the same linear system

for their prices. The measurement (`fuel_clearing_step1/convergence.py`, Fig. 5) is Table 5.

	market solves	converged?	distinct active sets	changes, 2nd half
$w = 0$	23	yes	5	0
$w = 1$, rigid balance	202	no	6	33
$w = 1$, demand-anchored	60	yes	6	0

Table 5: The failure, watched rather than deduced. At $w = 0$ the set of tight constraints stops moving and stays stopped. At $w = 1$ it never stops.

The two sets it alternates between differ in exactly two cells:

$$\{\text{obligation tight : 31, growth limit tight : 31}\} \longleftrightarrow \{\text{obligation tight : 29, growth limit tight : 33}\}.$$

The mechanism, precisely. In two (region, year) cells the binding constraint trades places between *the obligation binds* and *the growth limit binds*. On one side the obligation is met by volume and λ^M is the cost of substituting one fuel for another, about 0.012 EUR/MJ. On the other the obligation has become physically unreachable and the only way to discharge it is to pay — so λ^M snaps to the buy-out, 0.30 EUR/MJ. The delivered price jumps 23–29 % when it does.

The volume is identical on both sides. It is pinned by the growth limit. Only the price differs. That is not a solver artefact: at the crossing point the value function has a genuine kink, relaxing the growth limit saves nothing (the obligation is already met) while tightening it costs the buy-out, so the right and left derivatives differ and ∂V is the whole interval $[0.012, 0.30]$ — a factor of 25. This is exactly the caveat of §2.2, met in practice.

So w decides whether the model couples to a well-defined function of its own state, not how volatile the price is. At $w = 0$ the delivered price is the average cost: a function of volumes, which move by 1–2 %. At $w = 1$ it contains λ^M , and at a kink λ^M is not a function of the state at all.

Two explanations were tested and **rejected**. The bad years sit next to a ReFuelEU step, so the staircase obligation looked responsible — but replacing it with a smooth ramp through the same points still fails: smoothing moves the boundary rather than removing it. And the measured loop gain, ≈ 0.32 , is far below 1; a gain that low cannot diverge, which is what ruled out “the prices are volatile” at the outset.

6.5 The fix: quantity from supply, price from demand

The diagnosis dictates the remedy. Where the supply curve is vertical — a growth limit exhausted, an obligation reachable only by buying out — the quantity is determined and the price is not. That is not a pathology; it is the ordinary situation in any capacity-constrained market, and every such market resolves it the same way: **the quantity comes from supply and the price comes from demand**. The programme was solving a supply curve at a fixed quantity and then being asked for a price the supply side does not contain.

So the balance is given a slope. Around the incoming demand $D_{r,t}$ and the price $p_{r,t}^0$ that demand was formed at, introduce $a_{r,t}$ (free) and replace (2) and (3) with

$$\sum_p q_{r,p,t} = D_{r,t} + a_{r,t}, \quad \sum_{p \in S_r} q_{r,p,t} + x_{r,t} \geq m_{r,t} (D_{r,t} + a_{r,t}), \quad (14)$$

adding to the objective the consumer surplus given up, to second order:

$$\sum_{r,t} \delta_t \left[-p_{r,t}^0 a_{r,t} + \frac{a_{r,t}^2}{2\beta_{r,t}} \right], \quad \beta_{r,t} = \eta \frac{D_{r,t}}{p_{r,t}^0} \geq 0. \quad (15)$$

The quadratic keeps the programme convex. Stationarity in $a_{r,t}$ then reads

$$\lambda_{r,t}^E + m_{r,t} \lambda_{r,t}^M = p_{r,t}^0 - \frac{a_{r,t}}{\beta_{r,t}} \quad (16)$$

— the delivered marginal price equals the inverse demand at the quantity served. The left-hand side is what the airline faces per MJ: the energy, plus the share m of obligation that MJ drags with it (§2.5). The identity holds to 5×10^{-11} in the kernel tests, and it is the entire mechanism. On a vertical stretch of supply, a moves until the price sits where demand wants it; the gap that opens between that price and the marginal production cost is a **scarcity rent**, which is what a capacity-constrained producer earns, not an error.

This is exact, not a relaxation. At a fixed point of the coupling loop the price the market returns *is* the price the demand was formed at, so $p^0 = \lambda^E + m\lambda^M$, hence $a = 0$ by (16), hence (14)–(15) reduce to (2)–(3) identically. The loop therefore converges to a solution of the *unregularised* programme or it does not converge at all. There is no fixed point at which the added term is still bending the answer. Measured: $|a|/D = 6 \times 10^{-11}$ at convergence.

And η does not have to be right. It sets how far the price moves per sweep and nothing else. Linearising the loop gives the requirement $\beta > \frac{1}{2}|dD/dp|$: over-stating the demand response is safe and merely slow, under-stating it by more than a factor of two overshoots.

Device	solves	converged?	$ a /D$ final	
none (rigid)	202	no	—	the failure
volume anchor alone, $\rho = 0.5$	202	no	—	cannot help: the volume is pinned
$\eta = 0.05$	202	no	2.8×10^{-3}	slope too small, overshoots
$\eta = 0.2$	202	no	1.5×10^{-9}	market settled, chain had not
$\eta = 0.5$	59	yes	6.5×10^{-11}	
$\eta = 1.0$	150	yes	2.5×10^{-10}	over-damped
$\eta = 3.0$	202	no	2.9×10^{-5}	over-damped past the iteration budget

Table 6: Ablation at $w = 1$, two regions, no acceleration. The fastest η is an estimate of the chain’s own elasticity of drop-in demand to the delivered fuel price: the iteration is fastest when the slope it is told matches the slope it faces.

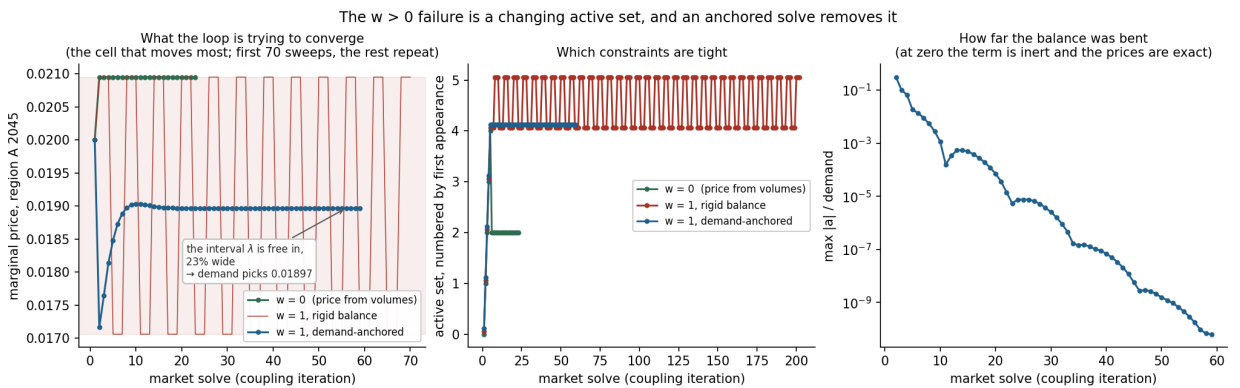


Figure 5: Left: the delivered price by sweep. Middle: the active set, numbered by first appearance — the value means nothing, only whether it changes. Right: how hard the anchor is pulling; it reaching zero is what certifies the answer is the unregularised one.

Two further consequences. First, a **negative result worth keeping**: anchoring the *volumes* instead — a proximal term $\frac{\rho}{2}\|q - \hat{q}\|^2$ — does not work, and cannot, because the primal is identical

on both sides of the kink. It moves λ^M across 0.6 % of the interval it is free in. It is retained as an optional, separately weighted device for degeneracies in the *primal*, off by default.

Second, all of this runs on **plain Gauss–Seidel with no acceleration**. The acceleration that was previously thought to help is gone from the coupled grid along with the claim.

Practical consequence: $w = 0$ was already solid. $w > 0$ is now a scenario dial rather than a research problem, at the cost of one parameter whose value affects the route and not the destination.

7 Policy cases

Five pathways — kerosene, HEFA from waste oil, HEFA from crop oil, alcohol-to-jet, e-fuel. *Kernel-level* (Table 2). Conclusions are about orderings and mechanisms; the numbers are illustrative.

7.1 Eligibility differing by region

One ReFuelEU-like region excludes crop feedstock; the other allows it; same headline obligation.

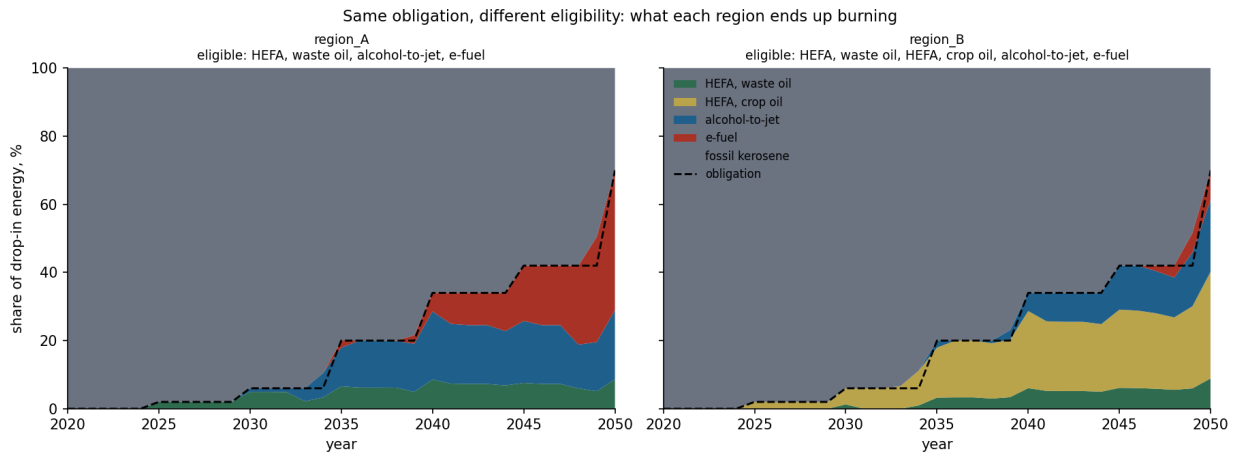


Figure 6: Same target, different portfolios. The coloured band is mandated fuel; the dashed staircase is the obligation, so colour above the line is anticipation.

Excluding a cheap feedstock is dearer every year except the last: +60 % on the compliance price in 2030, +139 % in 2040, and **−10 %** in 2050.

The mechanism is the growth limit. Denied the cheap option, region A builds e-fuel early and arrives at 2049 with a 30.9 % base; the permissive region coasts on crop oil and arrives with 5.0 %. When the obligation jumps 42 %→70 %, all four of the permissive region’s pathways hit their growth limits at once.

Cheap options early leave you worse placed for a steep step later — a result a share-allocation model cannot produce, because it has no notion of what was built when.

7.2 A sub-target within the target

ReFuelEU’s synthetic-fuel trajectory, on top of the headline obligation:

By 2050 e-fuel *exceeds* the 35 % target, because the forced early build-out leaves a bigger base to grow from.

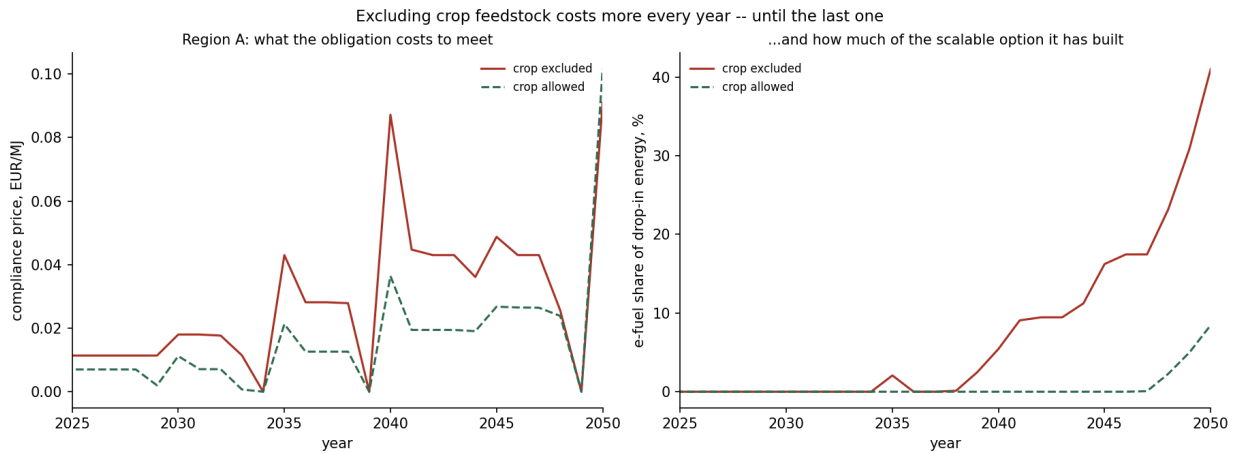


Figure 7: The cost, and the reason for it. Left: the exclusion is dearer in every year until the last; the sawtooth is each obligation step landing and the market catching up. Right: what it bought — a built-out e-fuel industry.

	2030	2040	2050
e-fuel, no sub-target	0.0 %	5.4 %	41.0 %
e-fuel, with sub-target	1.2 %	15.0 %	46.2 %
Main compliance price	0.01800 → 0.01721	0.08718 → 0.04103	0.09100 → 0.06908

A number that invites the wrong conclusion. The main compliance price falls in every year — by 53 % in 2040. That is *not* the sub-target making compliance cheaper: an extra constraint on the same programme can only make the optimum dearer, and it does. **Total production cost +10.7 %; fuel bill +5.1 %.** What falls is the price of *one instrument* because the other is now doing that work. Quoting the compliance price alone would show a policy paying for itself while it does the opposite.

7.3 A carbon tax, and nothing else

Every obligation switched off.

Below about 170 EUR/t a carbon tax on aviation fuel buys nothing at all. The mix stays 100 % kerosene and the intensity does not move from 88.6 gCO₂/MJ — it is pure revenue.

Then a cliff between 400 and 600, where kerosene goes from 67 % to zero and intensity falls 71.2 → 15.7. Not an artefact: kerosene's tax-inclusive cost ($0.0120 + 88.6\tau/10^6$) crosses e-fuel's ($0.0550 + 5.0\tau/10^6$) at $\tau = 514$ EUR/t, and the whole residual switches at once, because nothing distinguishes one litre of kerosene from another. **A tax moves one crossing point at a time, so its effect is a staircase, not a slope.**

7.4 The same destination by two instruments

Same 70 % sustainable, 2050	Production cost	Intensity	Mix
Obligation (crop excluded)	0.03427	36.5	41 % e-fuel, no crop
Carbon tax, 515 EUR/t	0.03056	41.3	33 % e-fuel, 18 % crop

The tax is **10.8 % cheaper** and **13 % dirtier**: it buys the cheapest sustainable fuel, the obligation's feedstock rule buys a cleaner one.

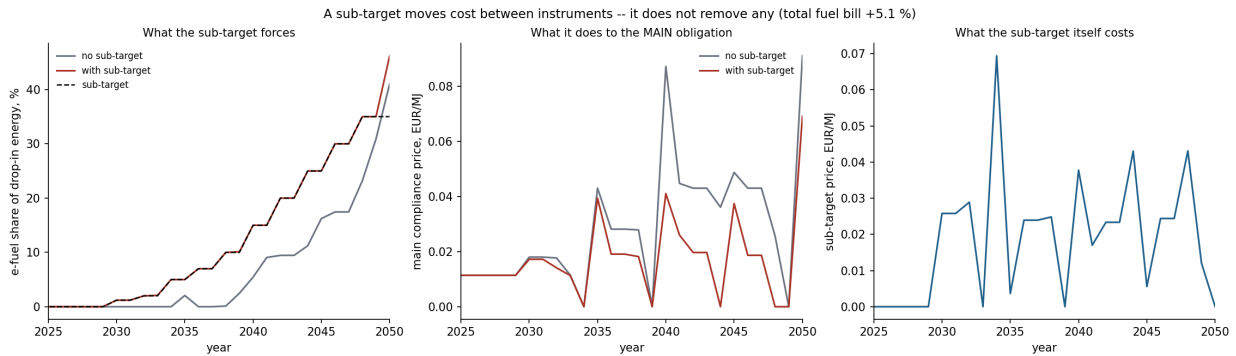


Figure 8: Left: the sub-target tracked exactly, then overshoot. Middle: the headline obligation gets cheaper throughout. Right: the sub-target's own price.

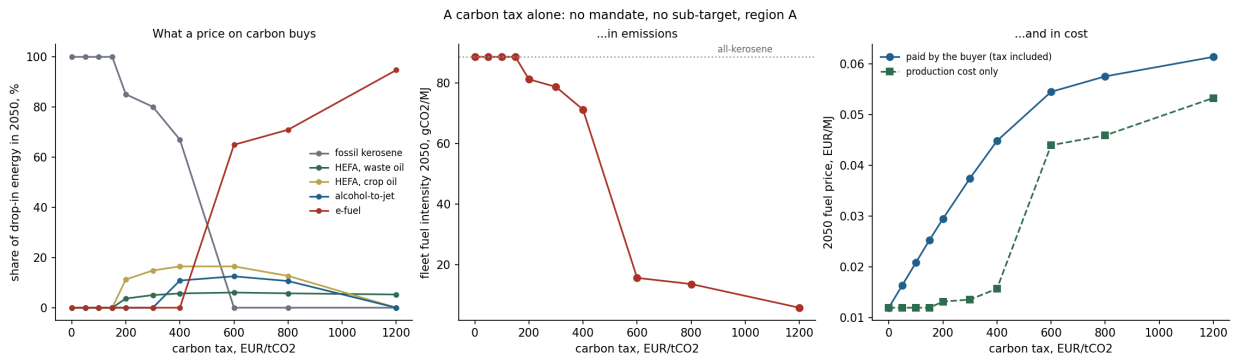


Figure 9: Nothing, then everything.

The environmental work is done by the eligibility rule, not by the volume target. A mandate written in volumes is not an emissions instrument. It becomes one only through the rules about what counts.

One confound: the tax case has no obligation, so no eligibility rule applies to it and crop oil is available. That is realistic — eligibility rules only exist inside mandates — but it makes this a *restricted mandate against an unrestricted tax*, not instrument against instrument at equal restriction.

8 Defects found, and one general lesson

Three defects of the *same family* were found, all of them producing plausible-looking numbers rather than errors.

1. **A price for an obligation that does not exist.** Where $m = 0$, constraint (3) is implied by $q, x \geq 0$ — redundant, so its multiplier is free. The solver reported up to 0.0112 EUR/MJ, roughly the whole cost of kerosene, in years with no obligation.
2. **An indeterminate split at a 100 % obligation.** There the obligation forces conventional supply to zero, so (2) and (3) have the same active rows and only their *sum* is determined. The solver split the identical problem three different ways, including a *negative* energy price.
3. **A jump at a binding/slack boundary** — §6.4. The same thing dynamically, and the one that stopped the coupled solve.

In every case a constraint stopped being independent and its multiplier became arbitrary. The first two are pinned to their meaningful KKT values. The third could not be pinned, because

the discontinuity is real and neither branch is more meaningful than the other — it needed information the cost side does not contain, which is what §6.5 supplies.

Two claims this report previously made and now withdraws

- **“Acceleration fixes the $w = 1$ failure.”** It was true of the one cell it was measured on. Run over the grid, Secant acceleration returned 3 cells of 12. It was a coincidence of trajectory, not a remedy: no sequence transformation can smooth a map that is genuinely discontinuous. The acceleration has been removed from the coupled grid.
- **“ $w > 0$ is a research problem, not a scenario dial.”** It was, until the failure was measured rather than inferred. It is now a dial (§6.5).

Both errors share a shape with the defects above, and with the four successive instrumentation attempts that each produced a confident and false reading of the coupled loop: *a measurement of one case, generalised*. The active-set signature exists so that this particular question can be answered by looking rather than by arguing.

A convex programme that *solves* is not a programme that is *right*. The saturation term was first written as $a q^{n+1}$ with $a = c\gamma/((n+1)K^n)$ — algebraically correct, numerically unusable: a carries K^{-n} and the coefficients spanned 35 orders of magnitude at $n = 16$. Half the parameter grid refused to solve; worse, *where it did solve it returned a suboptimal point*, 0.13% dearer, which showed up as a **3.4% error in the delivered price** — at the exact settings the whole sensitivity table was built on. The solver reported **optimal** both times.

Rewriting it in the utilisation it is actually a function of, $[c\gamma K/(n+1)](q/K)^{n+1}$, made all 36 cells solve. The only thing that caught the wrong answer was scoring the returned point against the objective *independently of the solver*.

9 Limitations

- **Regions do not interact.** Every constraint is per region and the objective is a plain sum, so solving jointly and separately agree to solver noise. There is no **leakage**: a stricter region cannot import compliance from a looser one. This is the single most important missing mechanism for policy work.
- **$w > 0$ needs a demand slope to converge** (§6.5). The value chosen does not affect the answer, but it is one more thing a user can set to a number that makes the run take 200 sweeps instead of 60. It should eventually be derived from the chain’s own elasticity rather than configured.
- **The elastic balance is a local linearisation.** It is exact at the fixed point and only there; away from it the demand curve it prices against is a straight line through one measured point. That is enough to select a price and not enough to be a demand model.
- **The buy-out payment reaches no one** (§2.5).
- **No learning curve** anywhere in AeroMAPS, so the cost of entry is a constant and every price dynamic comes from the model’s own scarcity terms.
- **Bottom-up costs are excluded “for now”**, and not for the usual reason. A vintage *mix* effect makes average cost *fall* as volume rises, which makes the cost integral concave and the duals stop being prices. A loop is not the obstacle; the sign is.
- **2050 is the horizon** and also the steepest step, so results turning on “what is built by the end” are partly a terminal condition.
- **Emission factors and four of the five costs are assumed.**

10 Where this goes next

1. **Inter-regional trade**, which is what the global (non-namespaced) discipline was built for. At step 1 it is global in plumbing, not economics.
2. **Derive the demand slope** from the chain instead of configuring it, so $w > 0$ has no free parameter at all. The convergence rate already measures it: the fastest η on the bench is an estimate of the true elasticity (Table 6).
3. **Pass the buy-out payment through** to the cost chain.
4. **Make `cvxpy` a proper optional dependency** before the mode ships.